# Sahaj Bhadja

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## Summary

Software engineer with 2+ years of experience building **full-stack product systems** across **TypeScript, React, Python, SQL, and cloud-native services**, shipping responsive UIs, APIs, and infrastructure-aware workflows with strong interest in **Kubernetes**, distributed systems, and platform engineering.

## Education

**University of Southern California** **Los Angeles, CA**
*Master of Science in Computer Science* *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

**PDEU**
*B.Tech. in Computer Science and Engineering* *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

## Experience

**USC Dornsife** **LA, California**
*Software Engineer* *September 2025 – Present*
- Built full-stack product infrastructure across **FastAPI**, **Azure SQL**, **PostgreSQL**, and cloud storage for a multi-tenant platform serving 500+ users, reducing dashboard latency from 1.2s to 350ms through schema and query optimization.
- Implemented fault-tolerant background-job and integration workflows for 5K+ monthly tasks with validation, reconciliation checks, and CI/CD-backed releases, improving throughput 3x while sustaining 99%+ production reliability.

**Grad8** **LA, California**
*Software Engineer Intern* *June 2025 – August 2025*
- Developed user-facing product workflows with **Flask** + **React**, secure file ingestion, and modular APIs, handling 10K+ daily grading requests for 150+ instructors in a production environment.
- Built concurrent processing and release workflows on **AWS RDS** with CI/CD-backed deployments, cutting feedback turnaround time by 60% while maintaining stable API response under peak load.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Engineered microservice-based APIs and cross-platform data workflows using **Python**, OOP patterns, and **AWS Glue/S3**, improving response times by 30% and pipeline efficiency by 68% in a production codebase.
- Maintained 20+ unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, strengthening delivery quality, debugging practices, and collaborative code review cycles.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*
- Built a MERN-based operations platform with **REST APIs**, WebSockets, and microservices, supporting 8+ business workflows and 2M+ synchronized records in a user-facing system.
- Implemented modular auth, billing, and notification services with **Jenkins**, **Docker**, and Azure SQL-backed workflows, reducing deployment time by 50% and sustaining 95%+ successful build validation.

**LST Capitals** **India**
*Web Developer Intern* *November 2022 – January 2023*
- Built backend APIs and real-time synchronization flows for a service-booking platform across 50K+ records, covering booking, reservation tracking, pricing, and payment status in a customer-facing product.
- Designed end-to-end system workflows across 12+ backend integration points with Firebase and payment webhooks, aligning implementation details with client requirements and production support needs.

## Skills

**Languages**: TypeScript/JavaScript, Python, Java, SQL, C++
**Backend & APIs**: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP, Microservices, TCP/IP, mTLS Concepts
**Frontend**: React.js, Next.js, TailwindCSS, HTML/CSS, WebSockets, State Management, UI Performance Profiling
**Databases**: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Relational Data Modeling, Caching Concepts
**Cloud & Infrastructure**: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Git/GitHub, Jenkins, Infrastructure Automation
**AI/ML**: LLM Systems, AI Coding Assistants, Prompt Engineering, Claude/Cursor/Gemini Workflows, Intelligent Automation
**Engineering Practices**: CI/CD, Unit Testing, Integration Testing, Version Control, Debugging, Distributed Systems, Apache Airflow Concepts, Go, Agile

## Projects

**PiXMod** | *Next.js, TailwindCSS, MongoDB, Stripe, Transformers*
- Built an AI-powered SaaS product with a responsive **Next.js** interface, image-processing workflows, and scalable storage in MongoDB, increasing premium conversion by 18% while shipping polished user-facing experiences.

**Chat2SQL** | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-assisted SQL workspace across a **Java** + **TypeScript** codebase, supporting 10+ databases and accelerating schema exploration, query authoring, and developer-facing data workflows.

**FinNexus** | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a full-stack fintech product with real-time updates, secure transaction controls, and **Kubernetes**-backed deployment across a distributed stack supporting 50K+ users and 1M+ transactions.